Memento

Save/restore text editor snapshots
Season 3 · GoF · Beginner

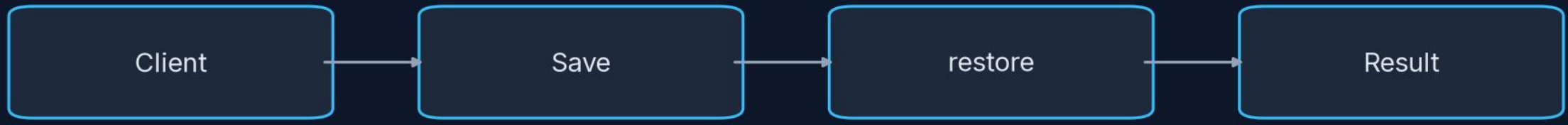
Today's Hook

- Problem: Save/restore text editor snapshots
- Naive pain cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- We will refactor to Memento with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Save/restore text editor snapshots
- Constraints: solve «Save/restore text editor snapshots» with Memento — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Save/restore text editor snapshots
- Success looks like: a

Structure



Memento: Save/restore text editor snapshots

Real-World Scenario

- Running example: Save/restore text editor snapshots
- Key types: DocMemento, Doc, History
- Happy-path stdout cue: a
- Failure-path stdout cue: Failure path is asserted in ProblemTest (injected fault / invalid input)

Naive Code

```java
package com.buildwithwaqar.patterns.gof.behavioral.memento.problem;

import java.util.*;

public final class NaiveDoc {
  public String text="";
  public String backup; // single slot only
  public void save(){ backup=text; }
  public void restore(){ text=backup; }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Save/restore text editor snapshots
- Without Memento, call sites absorb the wrong details.
- Failure mode to show on camera: Failure path is asserted in ProblemTest (injected fault / invalid input)
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.gof.behavioral.memento.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.function.*;

final class DocMemento { final String text; DocMemento(String t){ text=t; } }
final class Doc {
  private String text="";
  void write(String s){ text+=s; }
  String text(){ return text; }
  DocMemento save(){ return new DocMemento(text); }
  void restore(DocMemento m){ text=m.text; }
}
final class History {
  private final Deque<DocMemento> stack=new ArrayDeque<>();
  void push(DocMemento m){ stack.push(m); }
  DocMemento pop(){ if(stack.isEmpty()) throw new IllegalStateException("empty"); return stack.pop(); }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.gof.behavioral.memento.problem;
import java.util.*; import java.util.concurrent.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Memento: editor snapshots ===");
    System.out.println("--- naive ---");
var n=new NaiveDoc(); n.text="a"; n.save(); n.text="ab"; n.restore(); System.out.println(n.text);
System.out.println("--- memento ---");
var d=new Doc(); var h=new History(); d.write("Hello"); h.push(d.save()); d.write("!");
System.out.println(d.text()); d.restore(h.pop()); System.out.println("restored="+d.text());
  }
}
```

Walkthrough Output

```
=== Memento: editor snapshots ===
--- naive ---
a
--- memento ---
Hello!
restored=Hello
```

Step-by-Step Trace

- Match stdout to stages — look for: a
- Then the injected failure: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Re-run: java -cp … 20-memento ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Memento
// Prefer records + sealed where they shrink the demo.
// Problem: Save/restore text editor snapshots
```

Trade-offs

- Use Memento when «Save/restore text editor snapshots» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Memento in product language.

Related Patterns

- Name the pattern usually confused with Memento and separate the forces.
- Sketch the roles for «Save/restore text editor snapshots» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Save/restore text editor snapshots
- Naive pain → Memento boundary → Demo stdout.
- Remember the failure cue: Failure path is asserted in ProblemTest (injected fault / invalid input)
- Break a ProblemTest assertion, watch it fail, restore it.